

Derivation of the $q(F)$ Variational Update

Supervised Bayesian Matrix Factorization

Self-Contained Reference for `code/update_F.R`

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Abstract

This document derives the coordinate-ascent variational inference (CAVI) update for the biological factor matrix F in the Supervised Bayesian Matrix Factorization model. Unlike the L update (which combines genomics and survival), F appears *only* in the genomics likelihood. The update is a p -observation EBNM problem: each feature j provides one pseudo-observation x_j and pseudo-noise s_j .

A central algebraic property is derived in Section 4: the feature-specific noise precision τ_j *cancels* in the EBNM pseudo-observation $x_j = B_F[j]/A_F[j]$, but does *not* cancel in the pseudo-noise $s_j = 1/\sqrt{A_F[j]}$. This means all features receive the same point estimate (weighted OLS) but features with higher precision τ_j are subject to less EBNM shrinkage.

This is the third of four modular derivation documents ($q\beta \rightarrow qL \rightarrow qF$ (this) $\rightarrow q\tau$).

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1 Setup and Notation

1.1 Model Summary

The genomics likelihood is:

$$Y_{ij} = \sum_{k=1}^K l_{ik} f_{jk} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \tau_j^{-1}) \quad (1)$$

F does not appear in the Cox survival term. Therefore the entire $q(f_{jk})$ update is driven by the genomics likelihood alone.

1.2 Key Notation

Symbol	Definition	R variable
$\bar{l}_{ik}$	$\mathbb{E}_q[l_{ik}]$	EL[i, k]
$\bar{l}_{ik}^2$	$\mathbb{E}_q[l_{ik}^2]$	EL2[i, k]
$\bar{f}_{jk}$	$\mathbb{E}_q[f_{jk}]$	EF[j, k]
$\bar{f}_{jk}^2$	$\mathbb{E}_q[f_{jk}^2]$	EF2[j, k]
τ_j	Feature noise precision	Tau[j]
R_{ij}^{-k}	Partial residual (excl. factor k)	R_k[i, j]
Σ_{EL2_k}	$\sum_i \bar{l}_{ik}^2$ (scalar sum over samples)	sum_EL2_k

1.3 Partial Residual Matrix R^{-k}

$$R_{ij}^{-k} = Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'} \quad (2)$$

This quantity was defined and computed in the L update; the same `compute_R_k` function is reused here.

2 CAVI Objective for $q(f_{jk})$

The ELBO terms involving f_{jk} are:

$$\mathcal{F}[q(f_{jk})] \supset \mathbb{E}_q[\log P(Y | L, F, \tau)] + \mathbb{E}_{q_{f_k}} \left[\log \frac{g_{f_k}}{q_{f_k}} \right] \quad (3)$$

The genomics log-likelihood for feature j and patient i :

$$\log P(Y_{ij} | f_{jk}, \dots) \propto -\frac{\tau_j}{2} \mathbb{E}_q \left[(R_{ij}^{-k} - f_{jk} \bar{l}_{ik})^2 \right] \quad (4)$$

3 Derivation of A_F and B_F

Derivation Steps

Expanding the expectation in (4) and noting that $\mathbb{E}_q[l_{ik}^2] = \overline{l_{ik}^2}$:

$$\begin{aligned}\mathbb{E}_q[(R_{ij}^{-k} - f_{jk} l_{ik})^2] &= (R_{ij}^{-k})^2 - 2 f_{jk} R_{ij}^{-k} \mathbb{E}_q[l_{ik}] + f_{jk}^2 \mathbb{E}_q[l_{ik}^2] \\ &= (R_{ij}^{-k})^2 - 2 f_{jk} R_{ij}^{-k} \bar{l}_{ik} + f_{jk}^2 \overline{l_{ik}^2}\end{aligned}$$

Multiply by $-\tau_j/2$ and sum over all samples $i = 1, \dots, n$:

$$\sum_i -\frac{\tau_j}{2} \mathbb{E}_q[\dots] = -\underbrace{\frac{f_{jk}^2}{2} \tau_j \sum_i \overline{l_{ik}^2}}_{A_F[j]} + \underbrace{f_{jk} \tau_j \sum_i R_{ij}^{-k} \bar{l}_{ik}}_{B_F[j]} + \text{const} \quad (5)$$

Therefore:

$$A_F[j] = \tau_j \sum_i \overline{l_{ik}^2} = \tau_j \Sigma_{EL2_k} \quad (6)$$

$$B_F[j] = \tau_j \sum_i R_{ij}^{-k} \bar{l}_{ik} = \tau_j (R_k^\top \bar{l}_k)_j \quad (7)$$

where $\Sigma_{EL2_k} = \sum_i \overline{l_{ik}^2}$ is a *scalar*.

Code Mapping (update_F.R)

```
sum_EL2_k <- sum(EL2_k)           # scalar
A_F <- pmax(Tau * sum_EL2_k, A_floor) # p-vector [floored]
B_F <- Tau * as.vector(t(R_k) %*% EL_k) # p-vector
```

4 The τ Cancellation Property

This is a key structural property of the F update.

4.1 Cancellation in x_j

The EBNM pseudo-observation is:

$$x_j = \frac{B_F[j]}{A_F[j]} = \frac{\tau_j (R_k^\top \bar{l}_k)_j}{\tau_j \Sigma_{EL2_k}} \quad (8)$$

$$x_j = \frac{(R_k^\top \bar{l}_k)_j}{\Sigma_{EL2_k}} \quad \text{--- independent of } \tau_j \quad (9)$$

Interpretation: x_j is the weighted-least-squares estimate of f_{jk} that one would obtain if l_{ik} were known exactly (i.e., in the OLS limit where $\bar{l}_{ik}^2 = \overline{l_{ik}^2}$). The noise level τ_j does not affect the estimate itself.

4.2 No Cancellation in s_j

The EBNM pseudo-noise is:

$$s_j = \frac{1}{\sqrt{A_F[j]}} = \frac{1}{\sqrt{\tau_j \Sigma_{EL2_k}}} \quad (10)$$

τ_j does *not* cancel: features with higher precision τ_j (lower noise) receive smaller pseudo-noise s_j and hence *less* EBNM shrinkage.

Verification Check

Empirical check. With fixed R_k and EL_k , compute `update_F_k` twice: once with `Tau = rep(1, p)` and once with `Tau = rep(1000, p)`.

- `max|x(Tau=1) - x(Tau=1000)|` should be ≈ 0 (machine epsilon).
- `mean(s(Tau=1)) / mean(s(Tau=1000))` should be $\approx \sqrt{1000} \approx 31.6$.
- Posterior means should differ (less shrinkage at $\tau = 1000$).

This is verified in test T1.4 and T1.5 of `tests/test_update_F.R` and demonstrated in Demo 1 of `demos/demo_update_F.R`.

Correction Note

Correction R4 (from REVISED.tex): The EBNM subscript for the F update should be j (feature-indexed), not i . The update produces a p -vector of posterior means $\bar{f}_{jk}$ for $j = 1, \dots, p$. This was a typographical error in the original derivation document.

5 Error-in-Variables: Loading Uncertainty in A_F

The sum $\Sigma_{EL2_k} = \sum_i \bar{l}_{ik}^2$ uses the *full* second moment of each loading, not the squared posterior mean. This is the error-in-variables correction for uncertain loadings:

$$\sum_i \bar{l}_{ik}^2 = \sum_i (\text{Var}_q(l_{ik}) + \bar{l}_{ik}^2) = \underbrace{\sum_i \text{Var}_q(l_{ik})}_{\geq 0} + \sum_i \bar{l}_{ik}^2 \geq \sum_i \bar{l}_{ik}^2 \quad (11)$$

Using $\bar{l}_{ik}^2$ (pretending zero loading variance) would underestimate A_F , overestimating the signal-to-noise ratio and causing less shrinkage than appropriate.

6 EBNM Formulation

EBNM Problem

EBNM problem for $q(f_{\cdot k})$:

$$\text{EBNM}(x_j = B_F[j]/A_F[j], s_j = 1/\sqrt{A_F[j]}, g = g_{f_k}) \quad j = 1, \dots, p \quad (12)$$

Calling convention: `ebnm(x = B_F/A_F, s = 1/sqrt(A_F), prior_family = "point_normal")`

Returns: `posterior$mean` $\rightarrow \bar{f}_{jk}$, `posterior$sd` $\rightarrow \sigma_{jk}$
 Second moment: $\bar{f}_{jk}^2 = \sigma_{jk}^2 + \bar{f}_{jk}^2$

Code Mapping (`update_F.R`)

```
x_F <- B_F / A_F                # p-vector (tau-free!)
s_F <- 1.0 / sqrt(A_F)          # p-vector (tau-dependent)
res <- ebnm(x = x_F, s = s_F, prior_family = prior_family)
f_mean <- res$posterior$mean    # p-vector
f_sd <- res$posterior$sd        # p-vector
f_second <- f_sd^2 + f_mean^2   # p-vector
```

7 Gauss-Seidel Ordering in `update_F_all`

The wrapper `update_F_all` iterates over $k = 1, \dots, K$. After updating $EF[,k]$, the new values are used when computing R^{-k} for subsequent factors. Importantly, the *already-updated* $EL[,k]$ from the preceding L step is used here (Gauss-Seidel across the L-then-F sequence).

8 Verification Checks

Verification Check

- V1: OLS limit.** When $\bar{l}_{ik}^2 = \bar{l}_{ik}^2$ (zero loading variance), $x_j = (R_k^\top \bar{l}_k)_j / \sum_i \bar{l}_{ik}^2$, which is the OLS estimate of f_{jk} .
- V2: Null loading.** When $\bar{l}_{ik} = 0$ and $\bar{l}_{ik}^2 = 0$ for all i , then $\Sigma_{EL2_k} = 0$, so A_F hits the floor and $B_F = 0$, yielding $\bar{f}_{jk} = 0$.
- V3: Scalar Σ_{EL2_k} .** The loading second moments are summed over all n samples, giving a single scalar. Therefore $A_F[j] = \tau_j \times \text{const}$ — the only feature-variation in A_F comes from τ_j .
- V4: Second moment non-negativity.** $\bar{f}_{jk}^2 \geq \bar{f}_{jk}^2$ element-wise, with equality iff posterior variance is zero.

9 Code Mapping

Equation	Code	Type
$\Sigma_{EL2_k} = \sum_i \bar{l}_{ik}^2$	<code>sum(EL2_k)</code>	scalar
$A_F[j] = \tau_j \Sigma_{EL2_k}$	<code>Tau * sum_EL2_k</code>	p -vector
$B_F[j] = \tau_j (R_k^\top \bar{l}_k)_j$	<code>Tau * t(R_k) %*% EL_k</code>	p -vector
$x_j = B_F[j] / A_F[j]$	<code>B_F / A_F</code>	p -vector (τ -free)
$s_j = 1 / \sqrt{A_F[j]}$	<code>1/sqrt(A_F)</code>	p -vector (τ -dependent)

10 Algorithm Summary

- For $k = 1, \dots, K$ (Gauss-Seidel order):

- (a) Compute partial residual: $R^{-k} = Y - EL EF^\top + \text{outer}(EL[, k], EF[, k])$
- (b) Compute loading second-moment sum (scalar): $\Sigma_{EL2_k} = \sum_i \bar{l}_{ik}^2$
- (c) Precision (p -vector): $A_F[j] = \max(\tau_j \Sigma_{EL2_k}, \epsilon)$
- (d) Signal (p -vector): $B_F[j] = \tau_j (R_k^\top \bar{l}_k)_j$
- (e) EBNM pseudo-obs (τ -free): $x_j = B_F[j]/A_F[j]$
- (f) EBNM pseudo-noise (τ -dependent): $s_j = 1/\sqrt{A_F[j]}$
- (g) Solve EBNM: EBNM(x_j, s_j) over $j = 1, \dots, p$
- (h) Update $\bar{f}_{jk} \leftarrow$ posterior mean, $\overline{f_{jk}^2} \leftarrow \sigma_{jk}^2 + \bar{f}_{jk}^2$

A Notation Index

Symbol	Meaning	Dimensions
Y	Genomics data	$n \times p$
L	Patient loadings	$n \times K$
F	Biological factors	$p \times K$
τ_j	Feature noise precision	p -vector
R^{-k}	Partial residual (excl. k)	$n \times p$
Σ_{EL2_k}	$\sum_i \bar{l}_{ik}^2$	scalar
$A_F[j]$	EBNM precision for feature j	p -vector
$B_F[j]$	EBNM signal for feature j	p -vector
x_j	EBNM pseudo-obs (τ -free)	p -vector
s_j	EBNM pseudo-noise (τ -dependent)	p -vector